

Problem 1:

Prove that if R is a valuation ring, then R is normal.

Proof.

Let R be a valuation ring.

Let $x \in K = \text{Frac}(R)$ be integral over R . This means x is a root of a monic polynomial.

$x^n + a_{n-1}x^{n-1} + \cdots + a_0 = 0$ for $a_i \in R$.

Assume for contradiction that $x \notin R$. This implies that $x^{-1} \in R$ by definition of a valuation ring.

This implies that $x^n = -(a_{n-1}x^{n-1} + \cdots + a_0) \implies x = (-a_{n-1} + a_{n-1}x^{-1} + \cdots + x_0x^{-(n-1)})$.

This implies that $x \in R$.

Thus, R is normal. ■

Problem 2:

Let $(R, \mathfrak{m})$ be a noetherian local ring with residue field $k = R/\mathfrak{m}$.

(a)

Prove that $\mathfrak{m}$ is principal if and only if $\mathfrak{m}/\mathfrak{m}^2$ has dimension ≤ 1 as a k -vector space.

Proof.

Assume $\mathfrak{m} = (x)$ for some $x \in \mathfrak{m}$.

If $x = 0$ then $\mathfrak{m} = 0 \implies \mathfrak{m}/\mathfrak{m}^2 = 0 \implies$ it has dimension 0.

If $x \neq 0$ then this implies that $\bar{x}$ which is the image of x onto $\mathfrak{m}/\mathfrak{m}^2$ generates this ideal.

Let $\bar{y} \in \mathfrak{m}/\mathfrak{m}^2 \implies \pi(y) = \bar{y}$. Thus, $y = rx$ for some $r \in R$. Thus, $\pi(y) = \pi(r)\pi(x) = \bar{r}\bar{x}$.

Therefore, $\bar{y} = \bar{r}\bar{x}$. Thus, $\mathfrak{m}/\mathfrak{m}^2 = (\bar{x})$. Thus, this has dimension 1.

Assume $\mathfrak{m}/\mathfrak{m}^2$ has dimension ≤ 1 as a k -vector space.

If $\dim(\mathfrak{m}/\mathfrak{m}^2) = 0$

Then $\mathfrak{m}/\mathfrak{m}^2 = 0 \implies \mathfrak{m} = \mathfrak{m}^2$.

Since $\mathfrak{m}$ is finitely generated, by Nakayama's lemma we have that $\mathfrak{m} = 0 = (0)$.

If $\dim(\mathfrak{m}/\mathfrak{m}^2) = 1$

Then there exists some $x \in \mathfrak{m}$ s.t. $\bar{x}$ generates $\mathfrak{m}/\mathfrak{m}^2 = (\bar{x})$.

Consider (x)

By construction, the image of (x) in $\mathfrak{m}/\mathfrak{m}^2$ generate $\mathfrak{m}/\mathfrak{m}^2$.

Thus, $\mathfrak{m} = (x) + \mathfrak{m}^2$.

By Nakayama's lemma since $\mathfrak{m}$ is finitely generated since R is noetherian we get that $\mathfrak{m} = (x)$.

Thus, $\mathfrak{m}$ is principal in both cases. ■

(b)

Prove that R is a DVR if and only if R is a domain and $\mathfrak{m}/\mathfrak{m}^2$ has dimension 1 as a k -vector space.

Proof.

Assume R is a DVR. This implies R is a PID from shown in class. Thus, by (a) this implies that $\dim \mathfrak{m}/\mathfrak{m}^2 \leq 1$, but since $\mathfrak{m} \neq 0$ since R is not a field this implies that $\dim \mathfrak{m}/\mathfrak{m}^2 = 1$.

Also R is a domain since it is a PID.

Assume R is a domain and $\mathfrak{m}/\mathfrak{m}^2$ has dimension 1 as a k -vector space.

By (a) this implies that $\mathfrak{m} = (x)$ for some $x \in R$. But since $\mathfrak{m}/\mathfrak{m}^2$ has dimension one

this implies $\mathfrak{m} \neq 0$.

Shown in class this implies that R is a DVR. ■

(c)

Give a counter example where $\mathfrak{m}/\mathfrak{m}^2$ has dimension 1 as a k -vector space, but R is not a DVR.

Proof.

Consider $R = k[x]/(x^2)$ for field k .

R is noetherian and local with maximal $\mathfrak{m} = (\bar{x})$.

Consider $(\bar{x})/(\bar{x}^2) = (\bar{x})/(0) \cong (\bar{x})$. Thus, $\dim(\mathfrak{m}/\mathfrak{m}^2) = 1$.

But R is not a DVR since it is not a domain. ■

Problem 3:

Let k be a field and let $f \in k[x, y]$ be an irreducible polynomial of the form $f = \ell + g$ where $\ell = ax + by$ is the linear part and $g \in (x, y)^2$. Let $R = k[x, y]/(f)$, let $\mathfrak{m} = (\bar{x}, \bar{y}) \subset R$ be the maximal ideal generated by the images of x and y in R . Prove that the localization $R_{\mathfrak{m}}$ is a DVR if and only if $\ell \neq 0$.

Proof.

Since $k[x, y]$ is a finite type k algebra this implies that $\dim k[x, y] = \dim R + \text{codim}(f) \implies \dim R = 1$. This is because $\dim k[x, y] = 2$ and by the principal ideal theorem in a domain we have $\text{codim}(f) = 1$. ■

Problem 4:

Prove that if R is a DVR with maximal ideal $\mathfrak{m}$, then the $\mathfrak{m}$ -adic completion $\widehat{R}$ of R is a DVR.

Proof.

■ Assume R is a DVR with maximal ideal $\mathfrak{m}$. Let $\mathfrak{m} = (\pi)$. ■

Problem 5:

In this problem, we show that normal noetherian rings decompose as products of normal noetherian domains.

(a)

Let R be a noetherian ring such that $R_{\mathfrak{p}}$ is a domain for every $\mathfrak{p} \in \text{Spec}(R)$. Prove that there exist domains $R_1, \dots, R_n$ such that there is an isomorphism $R \cong R_1 \times \dots \times R_n$.

Proof.

Let $x \in \widehat{R}$. Thus, $x = (x_i)$ s.t. $x_i = x_{i+1} \pmod{\mathfrak{m}^i}$ for all i .

Show that $x = \sum_{i=0}^{\infty} a_i \pi^i$.

Base Case: $n = 1$.

$x_1 \in R/(\pi)$. Choose $a_0 \in R$ whose image is x_1 .

Inductive Step: Suppose for some $n \geq 1$ we have chosen $a_0, \dots, a_{n-1} \in R$ s.t.

$x_n \equiv \sum_{i=0}^{n-1} a_i \pi^i \pmod{\pi^n}$.

Consider $x_{n+1} \in R/(\pi^{n+1})$. It's image in $R/(\pi^n)$ is x_n , and the image of $\sum_{i=0}^{n-1} a_i \pi^i$ in $R/(\pi^n)$ is also x_n . ■